Curriculum Vitae

RESEARCH EXPERIENCE

Senior Research Scientist, Institute for Climate Modelling **2018 – Present**

- Lead a group of six researchers developing regional climate models for flood risk assessment.
- Secured £2.1M in competitive grant funding from national research councils.
- Built a Python and Fortran modelling pipeline running on a 2,000-core cluster.

Postdoctoral Researcher, University of Edinburgh **2014 – 2018**

- Developed statistical downscaling methods adopted by three European meteorological agencies.
- Supervised four MSc dissertations on hydrological modelling.

EDUCATION

PhD in Atmospheric Science, University of Reading, 2014

MSc in Applied Mathematics, Imperial College London, 2010

BSc in Physics, University of Manchester, 2009

PUBLICATIONS

1. Chen, A. et al. (2024). Regional flood risk under high-emission scenarios. Journal of Hydrology, 612, 128–141.
2. Chen, A. and Morris, J. (2023). Statistical downscaling for convective rainfall. Climate Dynamics, 60, 455–470.
3. Chen, A. (2022). Ensemble calibration for river basin forecasts. Water Resources Research, 58(4).

4. Chen, A., Patel, R. and Okafor, N. (2021). Uncertainty in compound flood events. Nature Climate Change, 11, 900–907.
5. Chen, A. (2020). A pipeline for kilometre-scale regional modelling. Geoscientific Model Development, 13, 2211–2230.
6. Chen, A. et al. (2019). Seasonal forecasts for reservoir operations. Hydrology and Earth System Sciences, 23, 77–94.
7. Chen, A. and Liu, H. (2018). Bias correction for daily precipitation. International Journal of Climatology, 38, 1201–1215.
8. Chen, A. (2017). Extreme value methods for urban drainage. Journal of Flood Risk Management, 10, 321–333.
9. Chen, A. et al. (2016). Multi-model ensembles for river discharge. Journal of Hydrometeorology, 17, 1709–1725.
10. Chen, A. (2015). Downscaling convective storms. Quarterly Journal of the Royal Meteorological Society, 141, 3010–3022.

GRANTS AND AWARDS

Natural Environment Research Council Standard Grant (2022) – £850,000

Royal Meteorological Society Early Career Award (2019)

European Geosciences Union Outstanding Paper Award (2017)

TEACHING

Lecturer, Introduction to Climate Science, University of Edinburgh (2015 – 2018)

Guest lecturer, Hydrological Modelling, University of Reading (2019 – present)

PRESENTATIONS

1. Invited talk, European Geosciences Union General Assembly, Vienna (2024).
2. Keynote, UK Flood Risk Conference, Birmingham (2023).

3. Invited talk, American Geophysical Union Fall Meeting, San Francisco (2022).
4. Contributed talk, International Conference on Climate Services, Copenhagen (2021).
5. Contributed talk, Royal Meteorological Society Annual Meeting, London (2019).

SKILLS

Python, Fortran, MATLAB, R, Statistics, Machine Learning, Linux, Git

REFERENCES

Available upon request.